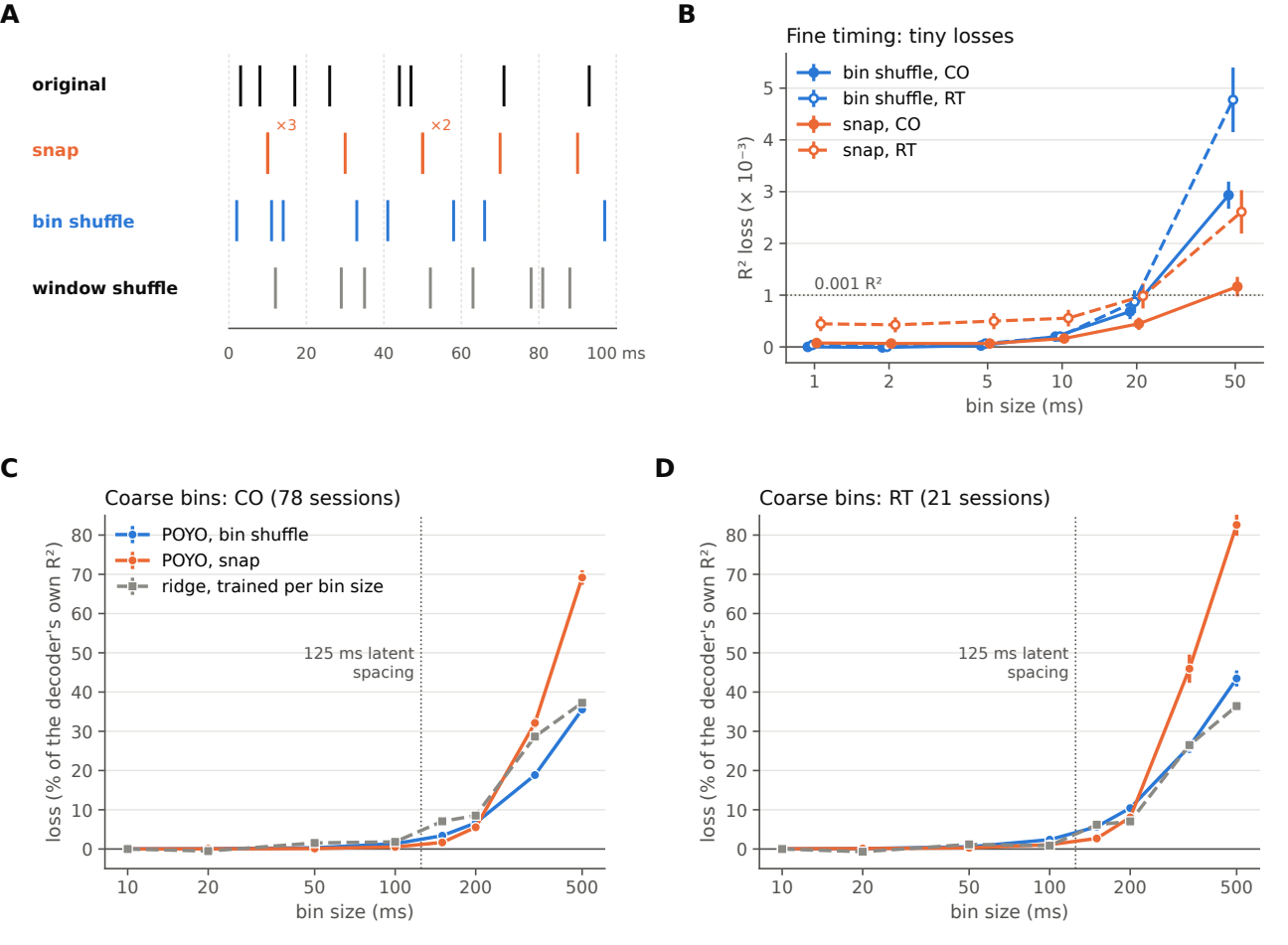
A
original
snap
bin shuffle
window shuffle
×3
×2
0
20
40
60
80
100 ms
B
Fine timing: tiny losses
bin shuffle, CO
bin shuffle, RT
snap, CO
snap, RT
R² loss (× 10⁻³)
0.001 R²
bin size (ms)
C
Coarse bins: CO (78 sessions)
POYO, bin shuffle
POYO, snap
ridge, trained per bin size
125 ms latent spacing
loss (% of the decoder's own R²)
bin size (ms)
D
Coarse bins: RT (21 sessions)
125 ms latent spacing
loss (% of the decoder's own R²)
bin size (ms)